

Proof. Norm non-negativity and degenerate cases. By definition $\|v\| = \sqrt{\langle v, v \rangle} \geq 0$ for all $v \in V$. If $x = 0$, then $\|x + y\| = \|y\| = \|0\| + \|y\|$; similarly if $y = 0$. Hence the inequality holds with equality in these cases. Henceforth assume $x \neq 0$ and $y \neq 0$.

Squaring the inequality. Both $\|x + y\|$ and $\|x\| + \|y\|$ are non-negative, so

$$\|x + y\| \leq \|x\| + \|y\| \iff \|x + y\|^2 \leq (\|x\| + \|y\|)^2. \quad (1)$$

Expansion of $\|x + y\|^2$. Using linearity of the inner product in the first argument and conjugate symmetry,

$$\|x + y\|^2 = \langle x + y, x + y \rangle \quad (1)$$

$$= \langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle \quad (2)$$

$$= \|x\|^2 + 2 \operatorname{Re} \langle x, y \rangle + \|y\|^2. \quad (2)$$

Expansion of $(\|x\| + \|y\|)^2$.

$$(\|x\| + \|y\|)^2 = \|x\|^2 + 2\|x\| \|y\| + \|y\|^2. \quad (3)$$

Cauchy–Schwarz bound. The Cauchy–Schwarz inequality states

$$|\langle x, y \rangle| \leq \|x\| \|y\|. \quad (4)$$

Since $\operatorname{Re} z \leq |z|$ for any $z \in \mathbb{C}$,

$$\operatorname{Re} \langle x, y \rangle \leq |\langle x, y \rangle| \leq \|x\| \|y\|. \quad (5)$$

Comparison of the two sides. Substituting the bound (5) into (3) yields

$$\|x + y\|^2 = \|x\|^2 + 2 \operatorname{Re} \langle x, y \rangle + \|y\|^2 \quad (3)$$

$$\leq \|x\|^2 + 2\|x\| \|y\| + \|y\|^2 = (\|x\| + \|y\|)^2, \quad (4)$$

which is precisely inequality (1).

Taking square roots. Both sides of (1) are non-negative, so taking the (non-negative) square root preserves the order:

$$\|x + y\| \leq \|x\| + \|y\|.$$

Equality condition. Equality in the chain above occurs iff

- $\operatorname{Re} \langle x, y \rangle = |\langle x, y \rangle|$ (so $\langle x, y \rangle$ is a non-negative real number), and
- equality holds in Cauchy–Schwarz, i.e. x and y are linearly dependent: there exists $\lambda \in \mathbb{C}$ with $y = \lambda x$ (or $x = 0$ or $y = 0$).

If $y = \lambda x$ with $x \neq 0$, then $\langle x, y \rangle = \lambda \langle x, x \rangle = \lambda \|x\|^2$. The condition $\operatorname{Re} \langle x, y \rangle = |\langle x, y \rangle|$ forces λ to be a real number and, because the real part equals the modulus, a non-negative one. Hence $\lambda \geq 0$.

Consequently,

$$\|x + y\| = \|x\| + \|y\| \iff \begin{cases} x = 0 \text{ or } y = 0, \\ \text{or } y = \lambda x \text{ with } \lambda \geq 0 \ (\lambda \in \mathbb{R}). \end{cases}$$

Thus the triangle inequality holds for all vectors in an inner-product space, and equality occurs exactly in the stated cases. $\square$ $\square$